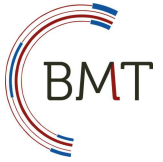


Universität Stuttgart

Institute of Biomedical Engineering



Models and Test Methods in Biomedical Engineering

Summer semester 2026

Practice – Test plan preparation + Test setup

Prof. Dr. Giorgio Cattaneo

Institute of Biomedical Engineering

University of Stuttgart

Seidenstraße 36, 70174 Stuttgart



Practice schedule

1	KW20	- Assignment description - Hydrogel model - Renal* artery segmentation (with 3D Slicer) - Mold design (with Siemens NX)	May 13
2	KW21	- Continuation of mold design (with Siemens NX) 3D printing preparation (with Preform)	May 20
	KW22	3D printing job to be sent per Email to Tobias Mörschel	May 29 at 5 pm
	KW23	Mold leak test	June 3
3	KW24	Test setup - Test plan preparation - Requirements definition - Outline of initial design concepts	June 10
4	KW25	- Short presentation (5 min): Test plan and design concepts - Test setup design finalization and 3D printing (with Bambulab)	June 17 Deadline: June 20 at 5 pm
5	KW26	- Model manufacturing - Assembly, initial operation, qualification	June 24
6	KW27	Testing and report	July 1
7	KW29	Team presentations	July 15

Flexibility and structural stability

User requirements (safety):

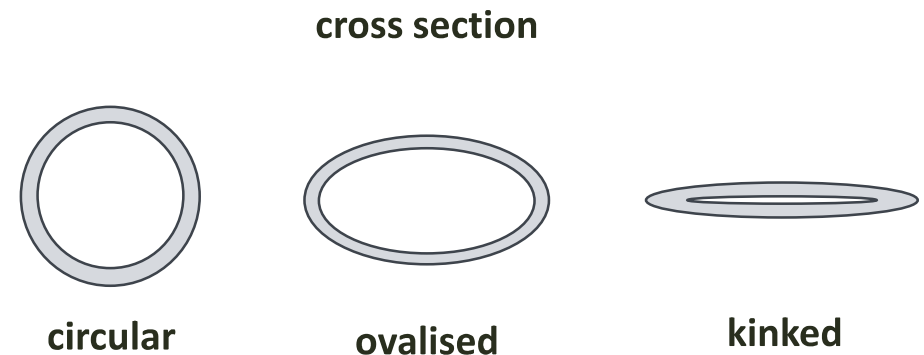
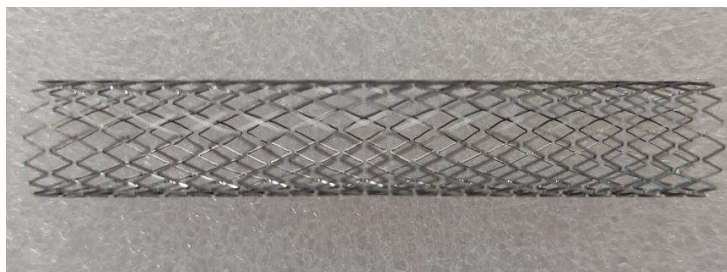
- Carotid stent must not disturb the hemodynamics
- Carotid stent must not damage the vessel while bending in the common carotid artery.

Engineering specification:

- Flexibility: Ability to **yield** under **bending force**
- Structural stability: Ability to undergo bending **without** cross-sectional **deformation (ovalization or kinking)**

Ovalization: When a circular or cylindrical structure bends, its circular cross-section deforms into an oval shape.

Kinking: sharp, localized bend



In vitro flexibility and structural stability test of carotid stents

Objective

To compare **two generations of stents** with regard to the **engineering specifications** by:

- Determining the **force** needed to bend the stent
- Determining the starting point of **ovalization**

in **two** different **conditions**: fully-expanded vs. implanted in a hydrogel model

Test material

- Nitinol stents, laser-cut with an 8 mm diameter and 50 mm length
 - Generation 1: rigid
 - Generation 2: flexible

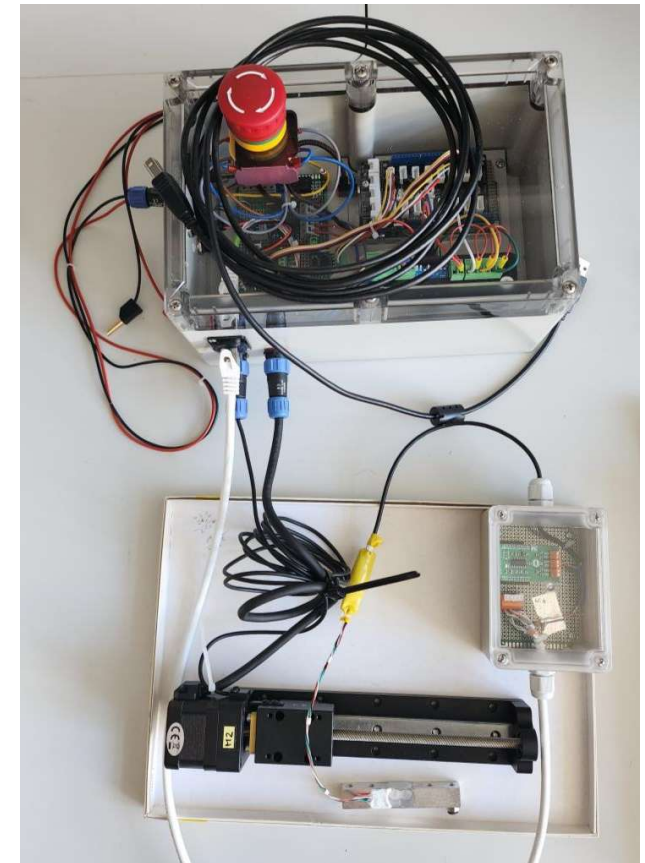
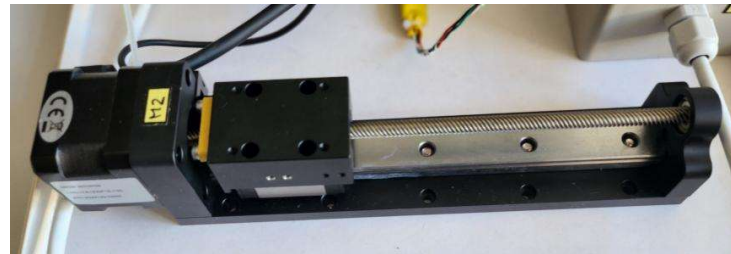
Acceptance criteria

To be defined (before tests are conducted!)

Test setup

Design, construction and initial operation of test setup

- Load cell: 1kg (if required 5kg)
 - Specification, data sheet
- Linear actuator: Stroke length: ~120 mm
 - Specification, data sheet
- Arduino control unit
 - Manual instructions
- Digital microscopes
 - Manual instructions
- Access to:
 - 3D printer – Bambulab
 - Computer/Laptop



Load cell

Load cell measures weight by converting force into an electrical signal.

Type:

- Single-ended shear beam load cell
- Strain gauge load cell

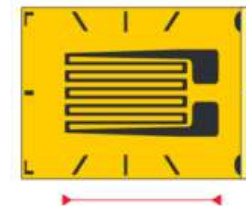
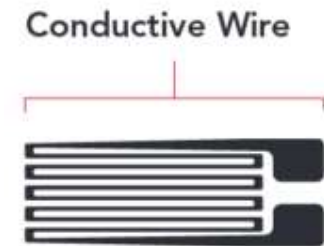
Strain gauge is an electrical element which can measure deformation of any material.

It is a very fine conductive wire or foil patterned as a grid

- When stretched $\rightarrow$ thinner $\rightarrow$ resistance increases
- When compressed $\rightarrow$ thicker $\rightarrow$ resistance decreases

Changes in resistance are very small but measurable.

Typically used in a **Wheatstone bridge** for accuracy.



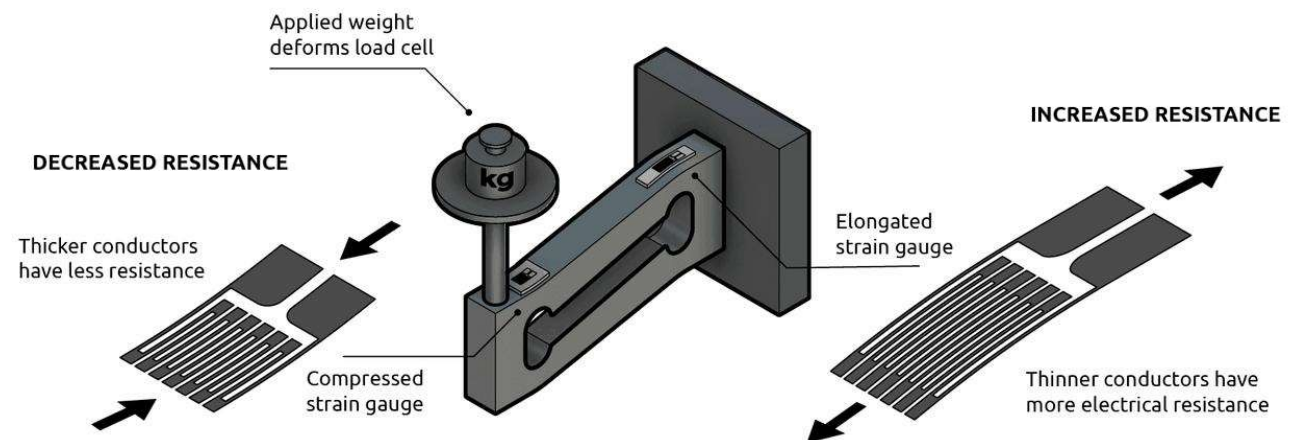
<https://www.anyload.com/strain-gauge/>

Load cell – Mode of operation

Strain gauges are bonded directly onto the metal surface, not visible from outside.

Usually **4 strain gauges** are used:

- 2 experience **tension**
- 2 experience **compression**



- Force applied → metal body of the load cell slightly deforms
- Strain gauges detect deformation → resistance changes

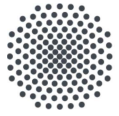
The strain gauges deform proportionally to the force applied.

- Wheatstone bridge converts resistance → small voltage signal
- Small signal → amplified, digitalized and read by a microcontroller.

<https://www.flintec.com/learn/weight-sensor/load-cell>

Practice schedule

1	KW20	- Assignment description - Hydrogel model - Renal* artery segmentation (with 3D Slicer) - Mold design (with Siemens NX)	May 13
2	KW21	- Continuation of mold design (with Siemens NX) 3D printing preparation (with Preform)	May 20
	KW22	3D printing job to be sent per Email to Tobias Mörschel	May 29 at 5 pm
	KW23	Mold leak test	June 3
3	KW24	Test setup - Test plan preparation - Requirements definition - Outline of initial design concepts	June 10
4	KW25	 Short presentation (5 min): Test plan and design concepts Test setup design finalization and 3D printing (with Bambu)	June 17  Deadline: June 20 at 5 pm
5	KW26	- Model manufacturing - Assembly, initial operation, qualification	June 24
6	KW27	Testing and report	July 1
7	KW29	Team presentations	July 15



Universität Stuttgart

Institute of Biomedical Engineering

Models and Test Methods in Biomedical Engineering

Summer semester 2026

Practice – Test plan preparation + Test setup

Prof. Dr. Giorgio Cattaneo

Institute of Biomedical Engineering

University of Stuttgart

Seidenstraße 36, 70174 Stuttgart

